

Bash-Scripting

1. Creating a Script

Create a directory to store the scripts:

```
Mkdir -p /hari/scripts
```

```
Cd /hari/scripts/
```

Create file:

```
vi script.sh
```

Add code:

```
#!/bin/bash  
echo "Welcome"
```

Give permission:

```
chmod 755 script.sh
```

Run:

```
./script.sh
```

```
bash script.sh
```

Screenshot:

```
[root@ip-172-31-32-82 scripts]# vi script.sh  
[root@ip-172-31-32-82 scripts]# chmod 755 script.sh  
[root@ip-172-31-32-82 scripts]# ./script.sh  
bash: ./script.sh: command not found  
[root@ip-172-31-32-82 scripts]# ./script.sh  
Welcome to hari tech  
[root@ip-172-31-32-82 scripts]# bash script.sh  
Welcome to hari tech  
[root@ip-172-31-32-82 scripts]# |
```

2. Script format

```
#!/bin/bash  
# purpose: Testing script format  
# Date: 2/7/2026  
# modification: 2/7/2026  
  
name='Srihari'  
job='Devops engineering'  
echo my name is $name  
echo im working as a $job|
```

```
[root@ip-172-31-32-82 scripts]# vi script.sh  
[root@ip-172-31-32-82 scripts]# ll  
total 8  
-rwxr-xr-x. 1 root root 40 Jul  2 09:59 script.sh  
-rwxr-xr-x. 1 root root 176 Jul  2 10:49 script.sh  
[root@ip-172-31-32-82 scripts]# bash script.sh  
my name is Srihari  
im working as a Devops engineering  
[root@ip-172-31-32-82 scripts]# |
```

3.Basic script

```
#!/bin/bash
pwd
ls
whoami
date
cal
touch a b c
```

```
[ec2-user@ip-172-31-32-82 ~]$ vi hari.sh
[ec2-user@ip-172-31-32-82 ~]$ bash hari.sh
/home/ec2-user
hari.sh test
ec2-user
Thu Jul  2 11:46:45 UTC 2026
      July 2026
Su Mo Tu We Th Fr Sa
        1  2  3  4
 5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 31

[ec2-user@ip-172-31-32-82 ~]$ |
```

4.Basic administration task

```
#!/bin/bash
top | head -5
df -h
free -m
uptime
iostat
```

```
[ec2-user@ip-172-31-32-82 ~]$ vi hari.sh
[ec2-user@ip-172-31-32-82 ~]$ bash hari.sh
top - 11:53:20 up 2:41, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 105 total, 1 running, 104 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 912.9 total, 510.7 free, 164.2 used, 237.9 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 616.9 avail Mem
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs         420M    0  420M   0% /dev
tmpfs            457M    0  457M   0% /dev/shm
tmpfs            183M  440K  183M   1% /run
efivarfs         128K  2.7K  121K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p1   8.0G  1.6G  6.4G  21% /
tmpfs            457M    0  457M   0% /tmp
/dev/nvme0n1p128 10M  1.3M  8.7M  13% /boot/efi
tmpfs            92M    0   92M   0% /run/user/1000
                total        used        free      shared  buff/cache   available
Mem:            912          164          510           0         237         616
Swap:            0           0           0
11:53:20 up 2:41, 1 user, load average: 0.00, 0.00, 0.00
Linux 6.18.35-68.129.amzn2023.x86_64 (ip-172-31-32-82.eu-north-1.compute.internal) 07/02/26
x86_64_ (2 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           0.10    0.00    0.05    0.01    0.01   99.83

Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
nvme0n1            1.05        16.02         43.56          0.00       155539       422880         0
```

Script 1:

```
#!/bin/bash
p=pwd
l=ls
wi=whoami
d=date
c='cal 2026'

echo run run variables task
echo
$p
$l
$ls
$wi
$d
$c
```

```
[ec2-user@ip-172-31-32-82 ~]$ vi hari.sh
[ec2-user@ip-172-31-32-82 ~]$ vi srihari.sh
[ec2-user@ip-172-31-32-82 ~]$ chmod 755 srihari.sh
[ec2-user@ip-172-31-32-82 ~]$ bash srihari.sh
run run variables task

/home/ec2-user
a b c hari.sh srihari.sh test
a b c hari.sh srihari.sh test
ec2-user
Thu Jul 2 12:02:27 UTC 2026

2026

January
Su Mo Tu We Th Fr Sa
    1 2 3
 4 5 6 7 8 9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

February
Su Mo Tu We Th Fr Sa
 1 2 3 4 5 6 7
 8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28

March
Su Mo Tu We Th Fr Sa
 1 2 3 4 5 6 7
 8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30 31

April
Su Mo Tu We Th Fr Sa
    1 2 3 4
 5 6 7 8 9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30

May
Su Mo Tu We Th Fr Sa
    1 2
 3 4 5 6 7 8 9
10 11 12 13 14 15 16
17 18 19 20 21 22 23
24 25 26 27 28 29 30
31

June
Su Mo Tu We Th Fr Sa
 1 2 3 4 5 6
 7 8 9 10 11 12 13
14 15 16 17 18 19 20
21 22 23 24 25 26 27
28 29 30

July
Su Mo Tu We Th Fr Sa
    1 2 3 4
 5 6 7 8 9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 31

August
Su Mo Tu We Th Fr Sa
    1
 2 3 4 5 6 7 8
 9 10 11 12 13 14 15
16 17 18 19 20 21 22
23 24 25 26 27 28 29
30 31

September
Su Mo Tu We Th Fr Sa
    1 2 3 4 5
 6 7 8 9 10 11 12
13 14 15 16 17 18 19
20 21 22 23 24 25 26
27 28 29 30

October
Su Mo Tu We Th Fr Sa
    1 2 3
 4 5 6 7 8 9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

November
Su Mo Tu We Th Fr Sa
 1 2 3 4 5 6 7
 8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30

December
Su Mo Tu We Th Fr Sa
    1 2 3 4 5
 6 7 8 9 10 11 12
13 14 15 16 17 18 19
20 21 22 23 24 25 26
27 28 29 30 31
```

Script 2:

```
#!/bin/bash
echo hello, my name is srihari
echo
echo what is your name?
read name
echo
echo hello $name
echo
```

```
[root@ip-172-31-32-82 ec2-user]# bash input.sh
hello, my name is srihari

what is your name?
mahesh

hello mahesh

[root@ip-172-31-32-82 ec2-user]#
```

Script 3:

```
#!/bin/bash
a=`hostname`
echo hello, my name is $a
echo
echo what is your name
read b
echo
echo hello $b
echo
```

```
[root@ip-172-31-32-82 ec2-user]# bash output.sh
hello, my name is ip-172-31-32-82.eu-north-1.compute.internal

what is your name
hari

hello hari

[root@ip-172-31-32-82 ec2-user]#
```

Script 4:

```
root@ip-172-31-32-82:/home/ec2-user
#!/bin/bash
# Example of defining variables

a='Imran'
b='Afzal'
c='Linux class'

echo My first name is $a
echo My surname is $b
echo My surname is $c
```

```
[root@ip-172-31-32-82 ec2-user]# vi variables.sh
[root@ip-172-31-32-82 ec2-user]# chmod 755 variables.sh
[root@ip-172-31-32-82 ec2-user]# bash variables.sh
My first name is Imran
My surname is Afzal
My surname is Linux class
```

Script 5:

```
root@ip-172-31-32-82:/home/ec2-user
#!/bin/bash
# Read user input

echo "What is your first name?"
read a
echo

echo "What is your last name?"
read b
echo

echo Hello $a $b

[root@ip-172-31-32-82 ec2-user]# bash readinput.sh
What is your first name?
srihari

What is your last name?
boora

Hello srihari boora
[root@ip-172-31-32-82 ec2-user]#
```

Script 6:

```
root@ip-172-31-32-82:/home/ec2-user
#!/bin/bash

echo "hello `whoami`"
echo
echo "today is `date`"
echo
echo "number of user login: `who | wc -l`"

~

[root@ip-172-31-32-82 ec2-user]# vi run
[root@ip-172-31-32-82 ec2-user]# chmod 755 run
[root@ip-172-31-32-82 ec2-user]# bash run
hello root

today is Thu Jul  2 13:07:25 UTC 2026

number of user login: 2
```

Rename task:

```
root@ip-172-31-32-82:/home/ec2-user
#!/bin/bash
# This script will rename a file

echo "Enter the file name to be renamed"
read oldfilename

echo "Enter the new file name"
read newfilename

mv $oldfilename $newfilename
echo The file has been renamed as $newfilename

root@ip-172-31-32-82:/home/ec2-user
[root@ip-172-31-32-82 ec2-user]# vi task
[root@ip-172-31-32-82 ec2-user]# chmod 755 task
[root@ip-172-31-32-82 ec2-user]# bash task
Enter the file name to be renamed
task
Enter the new file name
task_compeleted
The file has been renamed as task_compeleted
[root@ip-172-31-32-82 ec2-user]#
```